

Solution Requirements

Date	10 JULY 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the application. API keys (Gemini) are stored securely using environment variables and are not exposed to users.	High
2	Authorization levels	The system allows only authorized access to model files and configuration resources. General users can only perform emotion prediction.	Medium

3	External interfaces	Integrates with Google Gemini API for AI-generated learning guidance, Hugging Face Transformers, Streamlit UI, and TensorFlow/PyTorch models.	High
4	Transactions processing	Accepts user text, preprocesses it, performs emotion detection using BiLSTM and Fine-tuned BERT, and generates personalized learning support.	High
5	Reporting	Displays detected emotion, confidence score, AI-generated recommendation, and stores prediction history in CSV format for analysis	Medium
6	Business rules, etc.	User input must not be empty. The detected emotion is determined using the fusion of BiLSTM and BERT predictions. Personalized recommendations are generated only after successful prediction.	High
7	Compliance to laws or regulations	User inputs are processed securely. API keys remain confidential. No personally identifiable information is permanently stored.	Medium
8	<i>Other</i>	Supports multiple emotions (Joy, Sadness, Anger, Fear, Love, Surprise, Neutral) and provides learning assistance tailored to the detected emotion.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should provide predictions and AI responses with minimal delay	Emotion prediction within 2–5 seconds .
2	Scalability	The application should support future integration of additional emotions, datasets, and AI models without major redesign.	Easily extensible architecture supporting additional models.
3	Security & Data Privacy	Protect API keys using environment variables and ensure secure handling of user input.	API keys hidden, no sensitive data exposed, secure configuration.
4	Reliability & Availability	The application should provide consistent predictions with minimal downtime	99% uptime during deployment
5	Usability & Accessibility	Simple, intuitive Streamlit interface allowing users to enter text and receive recommendations easily.	User-friendly interface requiring minimal training
6	<i>Other</i>	Modular code structure with separate preprocessing, prediction, and recommendation modules to simplify maintenance and deployment.	Compatible with Windows, Linux, and cloud deployment platforms